

ATTACHMENT (1)
TECHNICAL ORDERING DATA (TOD)

CONTRACT No: N0010425RX19780
ITEM: Pipe; Seamless 1IN

SECTION B - SUPPLIES/SERVICES:

NATIONAL STOCK NUMBER (NSN): 1H 4710-01-599-7071 X2

ITEM NAME: PIPE, 1 IN, SMLS, NI-CU

PART NUMBER: MIL-DTL-23520, 1.00NPS, SCH40

QUANTITY: 6,372 INCHES

SPECIAL CHARACTERISTICS: NPS 1 INCH, SCH 40, Seamless, LEVEL 1

SECTION C - DESCRIPTION/SPECIFICATIONS:

C.1. Manufacture per the attached Individual Repair Part Ordering Data (IRPOD), as modified by this contract.

C.1.1. The order of precedence for any conflicting documentation/requirements shall be as follows:

- (a) The contract (including attachments, modifications, amendments, etc.).
- (b) The IRPOD.
- (c) Drawings (if any) referenced by the IRPOD.
- (d) Specifications referenced by the IRPOD or drawing.

C.1.2. Manufacture in accordance with MIL-DTL-23520, except as modified herein. Ordering data for paragraph 6.2 of MIL-DTL-23520 is as follows:

- (a) "Tube and Pipe, Nickel Copper Alloy, Seamless, Air Melted (UNS N04400)", MIL-DTL-23520E (SH) of 30 August 2021.
- (b) Class IV (4), Finish C (Ground) or Finish D (Descaled), Size Schedule and Quantity to be as in Section B of the schedule.
- (c) Not Applicable
- (d) For 1/2" thru 2" NPS Sizes: furnish to outside diameter and wall thickness.
- (e) Class IV Lengths per Paragraph 3.3.2.3(b), except maximum lengths of 20 feet in lieu of 24 feet are preferable.
- (f) Material making requirements of Paragraph 3.6.2 apply along with the requirements of C.5 below.
- (g) Vendor to perform metal removal when applicable in lieu of purchaser
- (h) Liquid Penetrant Required

C.1.3. Ultrasonically inspect pipe per MIL-DTL-23520 and MIL-STD-2132.

- (a) The vendor shall notify the purchasing agent four (4) weeks in advance of ultrasonic testing to allow participation of Portsmouth Naval Shipyard personnel in witnessing the ultrasonic tests.

C.1.4. MIL-DTL-23520 paragraph 4.4.2.1 Clarification: Each finished pipe and tube, regardless of size, shall be ultrasonically examined over 100% of the Outside Diameter surface to assure compliance with the specified Wall Thickness requirements.

C.2. APPLICABLE DOCUMENTS:

C.2.1. Section 2.0 of the IRPOD lists applicable documents to be used (e.g., drawings, specifications, standards, etc.) and their required version level (e.g., revisions, amendments, change notices, interim changes, etc.). The following list of documents and version information takes precedence over version information (including the lack of any version information) identified by the IRPOD:

- (a) MIL-DTL-23520E, TUBE AND PIPE, NICKEL-COPPER ALLOY, SEAMLESS AIR MELTED (UNS N04400)

- (b) MIL-STD-792F CHANGE 1; IDENTIFICATION MARKING REQUIREMENTS FOR SPECIAL PURPOSE COMPONENTS
- (c) MIL-STD-129R; MARKING FOR SHIPMENT AND STORAGE
- (d) ASTM D3951-18; STANDARD PRACTICE FOR COMMERCIAL PACKAGING
- (e) MIL-STD-2132F; NONDESTRUCTIVE EXAMINATION REQUIREMENTS FOR SPECIAL APPLICATIONS

C.2.2. If no document version is identified (anywhere in the contract or referenced documents), then use the latest version identified in the Acquisition Streamlining and Standardization Information System (ASSIST) at the time of contract award; ASSIST information is available at the following website:

<https://quicksearch.dla.mil>. If no document version information is available in ASSIST, then any version of the document may be used provided its version is dated 1 November 1969 or later.

C.2.3. In any case, the vendor shall use only one version in its entirety (i.e., the vendor shall not use portions of different versions of a document).

C.3. DATA REQUIREMENTS:

C.3.1. Material Certification; See Exhibit A, Data Item No. A001.

C.3.2. Procedure Approvals; See Exhibit A, Data Item No. A002. The vendor shall provide, and obtain approval for, the procedures identified below prior to their use. If the vendor considers a required submittal inapplicable to this procurement, the vendor shall submit a statement justifying the exclusion of the required submittal.

- (a) ULTRASONIC TEST PROCEDURE - Procedure (per MIL-STD-2132 – for both soundness and thickness), including acceptance criteria (per MIL-DTL-23520), shall be submitted. Ultrasonic calibration standard certification shall also be submitted for review and approval (this may be submitted no later than two weeks prior to scheduling the ultrasonic inspection witness).
- (b) ALLOY IDENTITY – Each tube and pipe shall be alloy identity tested for the correct generic metallurgical group as defined in MIL-STD-2132. The test shall be performed after the final marking required by 3.6, unless the material passes directly from the alloy identity device to the marking device.
- (c) LIQUID PENETRANT TEST PROCEDURE – Procedure conforming to the applicable requirements of MIL-STD-2132 shall be used for liquid penetrant test examination.

C.4. CLEANLINESS REQUIREMENTS:

C.4.1. Cleanliness requirements shall be per Section 3.2 of the IRPOD and Paragraph 5.1 of MIL-STD-23520.

C.5. MATERIAL MARKING (LEVEL 1 MATERIAL):

C.5.1. Each piece shall be permanently marked on each end with the information of (a), (b), (c), and (d) below using the permanent marking methods of MIL-STD-792 or the permanent marking method specified in the material specification. If any marking specified below cannot be applied due to space limitation, a waiver request shall be submitted to the contracting officer.

- (a) The Material Designator/Code “NCF”.
- (b) The manufacturer’s name, trademark, or symbol.

- (c) The vendor traceability number traceable to all manufacture test and inspection information and to a unique set of physical, mechanical, and chemical test results which represent the material used to make the piece.
- (d) Any information required by the specification or drawing (if applicable) in addition to that required by (a), (b), and (c) above shall be marked either per the specification/drawing or on each piece at each end per the permanent marking methods authorized above. Any Material Designator/Code listed in the specification or drawing (if applicable) need not be marked on material, if it is different from that specified by (a) above.

C.5.2. The location of the markings shall be on the outer diameter on the end(s) of each piece. Die stamp markings shall not be used for permanently marking pipe or tube per MIL-STD-792.

C.5.3. If feasible, consolidate the placement of the permanent marking(s) to accommodate the marking of a nine digit final acceptance marking at Portsmouth Naval Shipyard. Do not compromise the legibility of the permanent markings to meet this request.

C.6. NAVSEA HQ C-2-0023 Mercury Control.

SECTION D - PRESERVATION/PACKAGING/PACKING:

D.1. Marking requirements for packaging shall be per Section 5.0 of the IRPOD.

D.2. Preservation, Packaging, and Packing shall be in accordance with ASTM D3951 and MIL-DTL-23520 paragraph 5.2, as modified below:

- (a) The material shall be shipped in wooden boxes. For 6-inch OD and larger pipe, the wooden boxes shall be constructed of 2-inch thick by 6-inch wide minimum planks nailed or screwed together. The exterior of the boxes shall be banded with metal straps.

D.3. The use of MIL-STD-2073-1 packaging codes are not required for this contract.

SECTION E - INSPECTION ACCEPTANCE:

E.1. Government inspection is required during manufacture and prior to shipment from this facility. Upon receipt of this order, promptly notify the Representative of the Government office who normally serves your facility so that appropriate planning for Government inspection can be accomplished. In the event a Representative of the Government office cannot be located, the purchasing agent should be notified promptly. Provide the location where Procurement Quality Assurance (PQA) of the supplies will be made along with the organization that will be performing PQA.

E.2. All non-conformances to contract requirements affecting product and/or documentation shall require procuring activity waiver approval to accept.

E.3. FAR 52.246-11, Higher-Level Contract Quality Requirements

Supplemented by:

NAVSEA HQ H-246-H001 Calibration System Requirements (NAVSEA)

NAVSEA HQ E-246-H020 Quality Management System Requirements (NAVSEA)

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NAVSEA HQ C-246-H002 Government use of Contractor's Inspection Equipment (NAVSEA)
(MIL-I-45208 or ISO 9001/9002 with STR-9000 Supplement A or B or C)

E.4. FAR 52.246-2 Inspection of Supplies-Fixed-Price (Aug 1996).